

concepts from features and neurons in Appendix (8,9). This dramatic difference persisted across model layers and disappeared in control experiments with randomized model weights. Notably, although the randomized models extract zero features corresponding to biological Swiss-Prot concepts (Figure 3), they have hundreds of features strongly associated with individual amino acid types (Appendix Figure 9).

3.4 Features Form Clusters Based on Shared Functional and Structural Roles

Clustering features by their dictionary vectors exposes groups with coherent biological functions and distinct specializations. Within a kinase-associated feature cluster, three features exhibited activation on binding site regions with subtle variations in their spatial preferences (Figure 4). One feature activated specifically on the conserved alpha helix and beta sheet (α_e , VI) preceding the catalytic loop, while others concentrated on distinct regions near the catalytic loop, as evidenced by their maximally activated examples in both structural and sequence representations (4b,c). Though their peak activation positions varied, all features maintained high activation levels (> 0.8) on their respective maximum examples across the cluster, suggesting they identify similar kinase subtypes.

Analysis of a beta barrel-associated cluster revealed a distinct generalization pattern. While all three features in the cluster were labeled as TonB-dependent receptor (TBDR) beta barrels, only one feature exhibited specificity for TBDR beta barrels, whereas the other two identified beta barrel structures more broadly, including TBDRs, as demonstrated by their maximally activating examples (4e). All three features exhibit F1 associations with TBDR beta barrels, yet they differ markedly in their specificity. f/1503 shows exceptional specificity (F1=.998), functioning as a true TBDR detector. The other features, though capable of identifying TBDR structures (high recall), also recognize various other beta barrel proteins (lower precision), resulting in lower but still significant F1 scores (.793, .611). This clustering demonstrates how the model’s embedding space captures the natural hierarchy of beta barrel structures, where specialized TBDR detectors and general beta barrel features maintain similar representations despite operating at different levels of structural specificity.

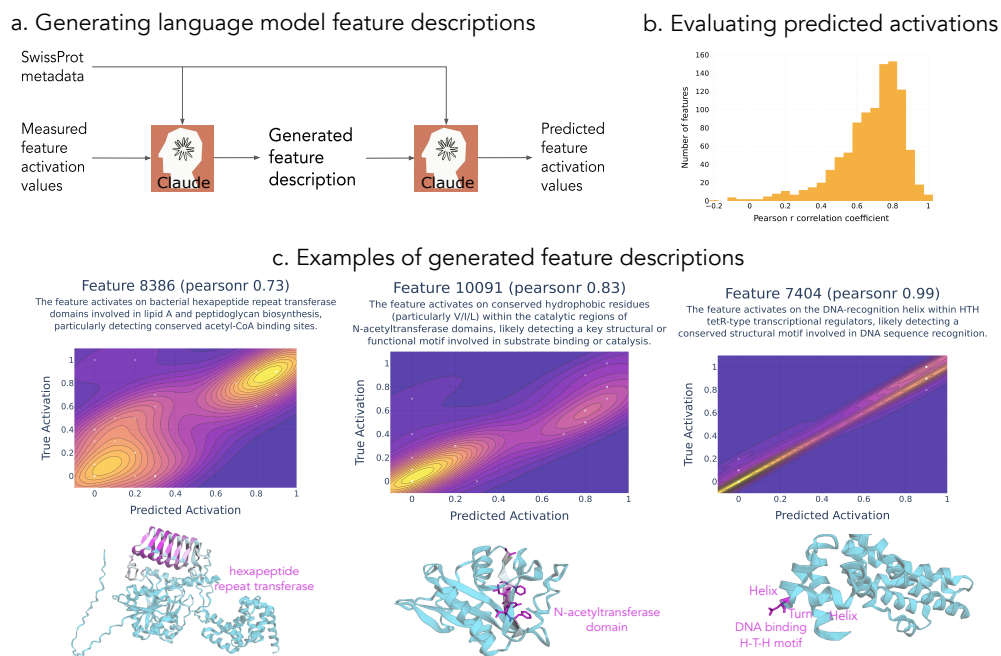


Figure 5: Language models can generate automatic feature descriptions for SAE features. (a) Workflow for generating and validating descriptions with Claude-3.5 Sonnet (new). (b) Comparing measured maximum activation values in proteins to predicted maximum activation values via Pearson r correlation across 1200 features. (c) Examples of generated feature descriptions and maximally activated proteins of each feature. Predicted activations quality visualized via kernel density estimation. The text is Claude’s description summary of each feature. Elements of description present in max examples annotated next to structures.

3.5 Large Language Models Can Generate Meaningful Feature Descriptions

To extend beyond Swiss-Prot concepts, which label less than 20% of features across layers (Appendix Figure 9a), we developed an automated pipeline using Claude to generate feature descriptions. By providing Claude-3.5 Sonnet (new) with the Swiss-Prot concept information including text information not applicable for classification, along with examples of 40 proteins with varying levels of maximum feature activation, we generate descriptions of what protein and amino acid characteristics activate the feature at different levels. As validation, the model generated descriptions and Swiss-Prot metadata are used to predict feature activation levels on separate proteins and showed high correlation with actual feature activation (median Pearson r correlation = 0.72) across diverse proteins. As shown in Figure 5, the descriptions accurately match specific highlighted protein features, with density plots revealing distinct clusters of correctly predicted high and low feature activations. The two examples with higher correlations identify specific conserved motifs directly tied to consistent protein structure and function, while the third example (f/8386) detects a structural motif (hexapeptide beta helix) that does not have Swiss-Prot annotations and appears across multiple functionally diverse proteins, potentially explaining the increased difficulty and lower performance at protein-level activation prediction. However, we note that across the distribution of tested features, there is only a weak dependence between a feature’s Swiss-Prot concept prediction performance and its LLM description accuracy (Pearson r = 0.11), suggesting that LLM-generated descriptions can effectively characterize many protein features regardless of whether they are easily categorized by annotations (Appendix Figure 10).

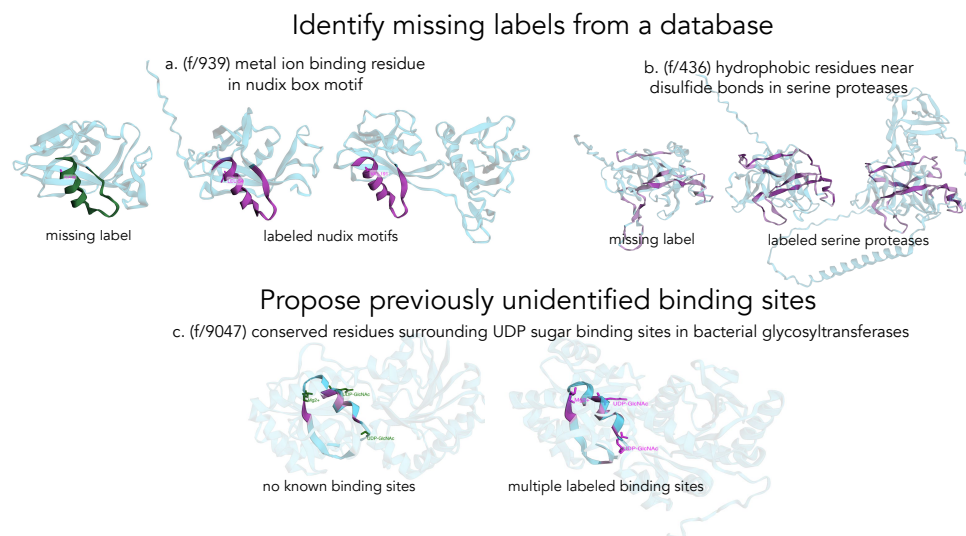


Figure 6: Feature activation patterns can be used to identify missing and new protein annotations. (a) f/939 identifies missing motif annotation for Nudix box. It activates on a single amino acid in conserved position which is labeled in each structure. Right two proteins are examples with high activations and pink coloring on the region with Nudix label. Left protein (B2GFH1), which does not have a Nudix motif annotation in Swiss-Prot, has implied feature activation highlighted in green. The presence of a Nudix motif somewhere in this protein is confirmed by InterPro. (b) f/436 identifies missing domain annotation for peptidase S1. It activates on a span of 80 amino acids with shared structural pattern. Right two proteins have high activation on confirmed peptidase S1 domains, with higher activation highlighted in pink. Left protein, which does not have peptidase S1 domain annotation in Swiss-Prot, is likely a missing annotation. The presence of an S1 domain somewhere in this protein is confirmed by InterPro. (c) f/9046 suggests missing binding site annotations for UDP-N-acetyl- α -D-glucosamine (UDP-GlcNAc) and Mg^{2+} within bacterial glycosyltransferases. In both structures, higher activation indicated with darker pink. Right protein has Swiss-Prot annotations for binding sites labeled in pink. Left protein, which has no known binding site annotations, but does have glycosyltransferase activity, has implied binding site annotations labeled in green

3.6 Feature Activations Identify Missing and Novel Protein Annotations

Analysis of features with strong concept associations revealed cases where "false positive" activations actually indicated missing database annotations. For example, f/939 highlights a single conserved amino acid in a Nudix box motif. We found one example that has high activation of 939 but no Swiss-Prot label, while every other highly activated protein does have the annotation. As seen in Figure 6, the region surrounding this activated position greatly resembles the labeled Nudix motifs. This feature is likely identifying a missing label, and if we examine the InterPro [24] entry for this

gene, we find reference to a Nudix motif, confirming this assumption. Similarly, f/436 can be used to identify Peptidase S1 domains with missing labels in Swiss-Prot despite positive identification in other external databases and Feature 9046 can be used to suggest the location of UDP and Mg2+ binding sites. The LLM-generated description of f/9046 emphasizes that it identifies UDP-dependent glycosyltransferases, however the majority of activated proteins only have text-based evidence of this, and no labeled annotations for binding sites. Using the binding site annotations from one protein, which each occur on amino acids adjacent to the activated positions, we can suggest potential annotations for the currently unlabeled proteins, although these would require further efforts to validate.

3.7 Protein Sequence Generation Can be Steered by Activating Interpretable Features

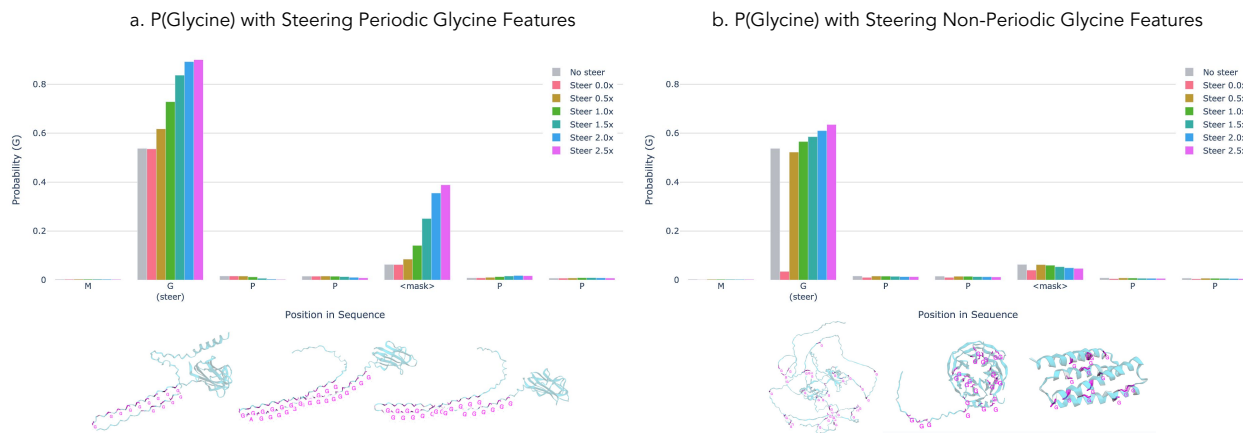


Figure 7: Steering feature activation on single amino acid additionally influences protein generation for nearby amino acids in interpretable way for group of periodic glycine features. Steering activation of features that activate on glycines in periodic repeats of GXX in collagen-like regions by only steering one G and measuring impact on nearby positions. (a) Steering periodic glycine features. The result of steering all 3 periodic features on the predicted probability values for glycine (G) at each position in the sequence MGPP<mask>PP. The feature steering was only applied to the unmasked G, not to the masked token or any other positions. Steer amount corresponds to the value the feature was clamped to where 1x is the maximum observed activation value of this feature observed. Maximally activating examples for the 3 Periodic features (f/4616, f/4970, f/10003) shown below (b) Steering non-periodic glycine features. Steering effects and maximally activated examples as in (a) but features selected to have the highest F1 score for glycine (f/6581, f/781, f/5381).

To validate that features capture causally meaningful patterns, we performed targeted interventions in the model’s predictions. Unlike in language models where features can be meaningfully steered by clamping their values across entire sequences, protein features rarely maintain consistent activation across domains. This makes it crucial to test whether localized feature manipulation can still influence model behavior at a distance. While many features capture complex biological concepts, quantitatively demonstrating specific interventions’ effects is challenging for patterns lacking clear sequence-based validation. We therefore focused on a simple, measurable example: showing how activating specific features can steer glycine predictions in periodic patterns.

We tested three features that activate on periodic glycine repeats (GXXGXX) within collagen-like domains. Given a sequence with a glycine followed by a mask three positions later, increasing these features’ activation values on the first glycine position increased the probability of glycine at both that position and the masked position - consistent with the expected periodic pattern (Figure 7a). Importantly, these features were originally identified by their characteristic activation pattern occurring every third amino acid in natural sequences. The fact that activating them on just a single glycine position - rather than their standard periodic distribution - still produced interpretable effects demonstrates the robustness of their learned patterns. As expected, highly glycine-specific features (F1 scores: .995, .990, .86) only influenced the directly modified position. In contrast, the periodic glycine features demonstrated a more sophisticated capability: they successfully steered predictions for unmodified positions, even propagating this effect for multiple subsequent periodic repeats with diminishing intensity (Figure 13), revealing their capture of higher-order sequence patterns.

These results demonstrate that features activating on interpretable biological patterns can be capable of causally influence model behavior in predictable ways, even at positions beyond direct manipulation. However, further research